INT303: Networking Fundamentals – Lab 1

Lab 1: Understanding Network Layers and TCP/IP Model using OWASP Broken Web Application IP

Objective:

In this lab, you will explore the OSI and TCP/IP models using the IP address of the OWASP Broken Web Application virtual machine. This practical exercise will help you understand how data is transmitted across networks, the key protocols involved, and how to use essential network commands for troubleshooting and analysis.

Learning Outcomes:

By the end of this lab, students will:

- Understand the OSI and TCP/IP models and their functions.
- Use essential network tools to explore and analyze network traffic.
- Apply knowledge of TCP/IP layers in real-world scenarios using OWASP Broken Web Application IP.

Materials Needed:

- Linux-based system (Kali, Ubuntu, etc.)
- OWASP Broken Web Application VM (running and reachable on your network)
- Terminal access with networking tools
- IP address of the OWASP Broken Web Application (e.g., 192.168.X.X)

Lab Exercises:

Exercise 1: Understanding OSI and TCP/IP Models

1. **Task:** Review the OSI and TCP/IP models and their layers.
    - OSI Layers: Physical, Data Link, Network, Transport, Session, Presentation, Application
    - TCP/IP Layers: Link, Internet, Transport, Application
2. **Question:**
    - List the OSI model layers and describe the function of each.
    - Match each OSI layer with its corresponding TCP/IP layer.

Exercise 2: Pinging the OWASP Broken Web Application

3. **Task:** Use the `ping` command to check the reachability of the OWASP Broken Web Application using its IP address.
    - Command: `ping <OWASP_IP>` (replace `<OWASP_IP>` with the actual IP, e.g., `ping 192.168.56.101`)
4. **Question:**
    - What happens when you ping the OWASP application? Describe the process.
    - Which OSI layer does the `ping` command operate in? Explain.

Exercise 3: Tracing the Path to the OWASP Application

5. **Task:** Use the `traceroute` command to trace the route packets take to reach the OWASP Broken Web Application.
    - Command: `traceroute <OWASP_IP>` (e.g., `traceroute 192.168.56.101`)
6. **Question:**
    - How many hops did it take to reach the OWASP VM?
    - Describe the significance of each hop and what role traceroute plays in network troubleshooting.

Exercise 4: Viewing Active Connections to OWASP VM

7. **Task:** Use `netstat` to view active network connections between your system and the OWASP application.
    - Command: `netstat -an | grep <OWASP_IP>`

8. **Question:**
    - What connections do you see? Identify the source and destination IP addresses.
    - Explain how the Transport Layer (TCP/UDP) is involved in this communication.

Exercise 5: TCP vs. UDP

9. **Task:** Investigate the difference between TCP and UDP by scanning the OWASP Broken Web Application.
    - Run a TCP scan using `nmap`.
    - Command: `nmap -sT <OWASP_IP>`
    - Run a UDP scan.
    - Command: `nmap -sU <OWASP_IP>`
10. **Question:**

- What are the key differences between TCP and UDP in terms of reliability and speed?
- Based on the scan results, list which services on the OWASP application are using TCP and which are using UDP.

Exercise 6: Discovering MAC Addresses with ARP

11. **Task:** Use the `arp` command to view the MAC address of the OWASP Broken Web Application.

- Command: `arp -a | grep <OWASP_IP>`

12. **Question:**

- What is the MAC address associated with the OWASP VM's IP?
- Explain the significance of ARP in the Data Link Layer and how it contributes to successful communication.

Exercise 7: Capturing Network Traffic with Wireshark

13. **Task:** Use Wireshark or `tshark` to capture network packets between your machine and the OWASP Broken Web Application.

- Command: `wireshark` (to launch the GUI) or `tshark -i <interface> host <OWASP_IP>`

14. **Question:**

- Analyze the captured traffic. What protocols are in use?
- Can you identify the handshake process or other significant events in the captured packets?

Submission Requirements:

- Submit a detailed report including answers to the questions and the commands you used.
- Include screenshots where appropriate, particularly for commands like `ping`, `traceroute`, `netstat`, and packet captures from Wireshark.
- Discuss any issues or troubleshooting steps taken during the lab.

Reflection:

At the end of this lab, students will have hands-on experience with network layer interaction using real-world tools and commands, along with a better understanding of how the TCP/IP model facilitates communication.

INT303: Networking Fundamentals – Lab 2

Lab 2: Exploring Network Interfaces and Packet Transmission Using OWASP Broken Web Application IP

Objective:

In this lab, you will dive deeper into the concept of network interfaces and packet transmission by analyzing the network interface configuration on your system. You will explore how to capture and analyze network packets using various tools and understand the process of packet transmission to and from the OWASP Broken Web Application VM.

Learning Outcomes:

By the end of this lab, students will:

- Understand the role and configuration of network interfaces.
- Use packet capture tools to analyze network traffic and understand how data is transmitted.
- Gain experience in monitoring and troubleshooting network interfaces and traffic.

Materials Needed:

- Linux-based system (Kali, Ubuntu, etc.)
- OWASP Broken Web Application VM (running and reachable on your network)
- IP address of the OWASP Broken Web Application (e.g., 192.168.X.X)
- Wireshark or tshark for packet capture

Lab Exercises:

Exercise 1: Viewing and Configuring Network Interfaces

1. **Task:** View your network interfaces using the `ifconfig` or `ip` command.
    - Command: `ifconfig` or `ip addr show`
2. **Question:**
    - What network interfaces are available on your system?
    - Identify the IP address assigned to each interface.
    - Describe the difference between a loopback interface and an external network interface.

Exercise 2: Capturing Packets on a Specific Interface

3. **Task:** Use `tcpdump` or `tshark` to capture packets on your primary network interface.
    - Command: `tcpdump -i <interface>` (e.g., `tcpdump -i eth0`) or `tshark -i <interface>`
4. **Question:**
    - What kind of packets are being captured?
    - Are there any packets related to communication with the OWASP Broken Web Application VM?
    - Describe the role of this network interface in transmitting and receiving packets.

Exercise 3: Examining Network Statistics

5. **Task:** Use the `netstat` or `ss` command to view current network statistics and connections.
    - Command: `netstat -i` or `ss -s`
6. **Question:**
    - What is the current status of your network interfaces?
    - What active connections are visible?
    - Explain the significance of these statistics in monitoring network performance.

Exercise 4: Monitoring Network Traffic with Wireshark

7. **Task:** Use Wireshark to monitor network traffic on your primary interface and filter traffic going to and from the OWASP Broken Web Application VM.
    - Command: Open Wireshark, choose your network interface, and set the filter to `ip.addr == <OWASP_IP>` (replace `<OWASP_IP>` with the actual IP, e.g., `192.168.56.101`)
8. **Question:**
    - Analyze the packets going to and from the OWASP VM. What types of protocols are in use?
    - Can you identify any key packet details such as source/destination IP addresses, port numbers, and flags?
    - How does the TCP/IP model apply to the data captured?

Exercise 5: Packet Transmission Analysis

9. **Task:** Perform a ping or TCP connection request to the OWASP Broken Web Application IP and capture the packet details using `tcpdump` or Wireshark.
    - Command: `ping <OWASP_IP>` or `telnet <OWASP_IP> 80` (to initiate a simple connection)
    - Capture the traffic using `tcpdump -i <interface> host <OWASP_IP>`, or set the appropriate filter in Wireshark.
10. **Question:**

- What do you observe during the packet transmission process?
- Describe the handshake process or the round-trip of the packets for ping or TCP connection.
- Which layers of the OSI and TCP/IP models are involved in this transmission?

Exercise 6: Troubleshooting Network Interface Issues

11. **Task:** Simulate a network interface failure by disabling and then re-enabling an interface. Observe how this affects network connectivity to the OWASP Broken Web Application VM.

- Command: `sudo ifconfig <interface> down` and `sudo ifconfig <interface> up` (or use `ip` commands)

12. **Question:**

- What happens when you disable the network interface?
- How does your system respond when the interface is re-enabled?
- Explain how network administrators can use this knowledge for troubleshooting connectivity issues.

Exercise 7: Bandwidth Monitoring

13. **Task:** Use the `iftop` or `nload` command to monitor the bandwidth usage on your system while interacting with the OWASP Broken Web Application VM.

- Command: `sudo iftop -i <interface>` or `nload <interface>`

14. **Question:**

- What is the current bandwidth usage while communicating with the OWASP VM?
- Identify the impact of network traffic on your interface. Is there any traffic congestion?
- How does this help in monitoring network performance?

Exercise 8: Advanced Packet Capture Filters

15. **Task:** Create custom filters in Wireshark or `tcpdump` to capture only specific types of traffic, such as TCP or HTTP traffic, between your system and the OWASP VM.

- Example Filter for HTTP Traffic: `tcp port 80 and ip.addr == <OWASP_IP>`

16. **Question:**

- What is the significance of filtering specific traffic?
- How can advanced filters help network engineers diagnose and resolve issues?

Submission Requirements:

- Submit a report detailing the output of each exercise.
- Include screenshots for critical steps such as packet captures and bandwidth monitoring.
- Provide explanations for each observed behavior, particularly focusing on the significance of network interfaces and packet transmission.

Reflection:

By the end of this lab, students will have a practical understanding of network interfaces and packet transmission. They will be able to configure and monitor network interfaces, analyze packet transmission, and troubleshoot network connectivity using real-world tools.

INT303: Networking Fundamentals – Lab 3

Lab 3: TCP/IP Protocol Stack and Packet Inspection Using OWASP Broken Web Application IP

Objective:

In this lab, students will delve into the Transmission Control Protocol/Internet Protocol (TCP/IP) stack, dissecting and inspecting the various layers involved in packet transmission. Students will focus on understanding how data travels across a network, analyzing packets, and troubleshooting issues related to the TCP/IP model.

Learning Outcomes:

By the end of this lab, students will:

- Understand the fundamentals of the TCP/IP model.
- Be able to analyze packets at different layers (Application, Transport, Network, and Link).
- Gain hands-on experience in dissecting TCP/IP traffic using packet inspection tools.
- Learn how to apply practical troubleshooting techniques based on TCP/IP behavior.

Materials Needed:

- Linux-based system (Kali, Ubuntu, etc.)
- OWASP Broken Web Application VM (running and reachable on your network)
- IP address of the OWASP Broken Web Application (e.g., 192.168.X.X)
- Wireshark or tshark for packet capture and analysis

Lab Exercises:

Exercise 1: Understanding the TCP/IP Model

1. **Task:** Briefly review the layers of the TCP/IP model (Application, Transport, Network, and Link layers). Understand the purpose and function of each layer in data transmission.
2. **Question:**
    - Explain the differences between the TCP/IP and OSI models.
    - Which layers of the TCP/IP model correspond to specific OSI layers?

Exercise 2: Capturing and Analyzing TCP Packets

3. **Task:** Initiate a connection to the OWASP Broken Web Application VM via HTTP or SSH. Capture TCP packets related to this connection using `tcpdump` or Wireshark.
    - Command: `tcpdump -i <interface> tcp and host <OWASP_IP>` or set a filter in Wireshark.
    - Example: `tcpdump -i eth0 tcp and host 192.168.56.101`
4. **Question:**
    - What happens during the TCP handshake (SYN, SYN-ACK, ACK)?
    - Identify and describe the flags used in TCP communication (SYN, ACK, FIN, etc.).
    - How does the TCP connection maintain reliability during transmission?

Exercise 3: Investigating IP Packets (Network Layer)

5. **Task:** Capture and analyze the IP packets from your connection to the OWASP Broken Web Application.
    - Command: Use Wireshark or `tcpdump` with the filter `ip.addr == <OWASP_IP>` to capture IP packets.
6. **Question:**
    - What fields can you see in the IP packet header (e.g., Source IP, Destination IP, TTL, etc.)?
    - What is the significance of each of these fields?

- How does IP routing work in this scenario? Are there any hops between your system and the OWASP VM?

Exercise 4: Application Layer Analysis (HTTP/SSH Traffic)

7. **Task:** If you're using HTTP, filter the packets to capture only HTTP traffic using Wireshark or `tcpdump`. If you're using SSH, capture and analyze the encrypted traffic.
   - Command for HTTP traffic: `tcpdump -i <interface> port 80 and host <OWASP_IP>`
   - Command for SSH traffic: `tcpdump -i <interface> port 22 and host <OWASP_IP>`
8. **Question:**
   - Analyze the HTTP packets. What information is available in the HTTP request and response?
   - For SSH traffic, what is the significance of encrypted packets? Can you analyze the payload?
   - How does the application layer play a role in data exchange between your system and the OWASP VM?

Exercise 5: Error Handling in TCP/IP Transmission

9. **Task:** Simulate packet loss or network issues by introducing a delay or stopping the network connection momentarily. Capture how TCP/IP handles this issue.
   - Command: Temporarily disable your network interface using `ifconfig <interface> down` and then re-enable it.
10. **Question:**
    - What happens when packets are dropped or delayed?
    - How does TCP ensure data reliability in the presence of errors?
    - How do retransmissions and sequence numbers work in TCP to maintain a proper data flow?

Exercise 6: ICMP and Ping Inspection (Network Layer)

11. **Task:** Send a series of `ping` commands to the OWASP Broken Web Application VM and capture the ICMP packets.
    - Command: `ping <OWASP_IP>` and use Wireshark or `tcpdump` to capture ICMP packets with the filter `icmp`.
12. **Question:**
    - What are the key fields in an ICMP packet (e.g., Type, Code, Checksum)?
    - How does ICMP assist in diagnosing network connectivity issues?
    - What is the significance of TTL in both ICMP packets and general IP packets?

Exercise 7: Analyzing UDP Packets (Transport Layer)

13. **Task:** If available, generate some UDP traffic to the OWASP VM by using a simple application or UDP-based service. Capture and analyze the UDP packets.
    - Command: `tcpdump -i <interface> udp and host <OWASP_IP>`
14. **Question:**
    - Compare UDP with TCP. What are the major differences in packet structure and behavior?
    - Why does UDP not ensure reliability, and in what scenarios would you prefer UDP over TCP?
    - How does UDP manage data transmission without the need for acknowledgments or retransmissions?

Exercise 8: OS Detection via Nmap (Network and Transport Layers)

15. **Task:** Use `nmap` to perform OS detection on the OWASP Broken Web Application VM, analyzing how `nmap` identifies the operating system based on packet behavior.
    - Command: `nmap -O <OWASP_IP>`
16. **Question:**
    - How does `nmap` detect the OS based on the captured packets?
    - What packet characteristics (TTL, window size, etc.) help in identifying the OS?

- Explain why OS detection can be an important step in network analysis and vulnerability assessment.

Exercise 9: Analyzing ARP Traffic (Link Layer)

17. **Task:** Use `arp-scan` or `tcpdump` to capture ARP traffic in your local network, focusing on communication with the OWASP VM.
    - Command: `arp-scan -I <interface> <OWASP_IP>`
18. **Question:**
    - What information can you gather from ARP packets (e.g., MAC addresses)?
    - How does the ARP protocol function at the link layer?
    - What role does ARP play in the communication between your system and the OWASP VM?

Exercise 10: Troubleshooting Network Connectivity Using TCP/IP Knowledge

19. **Task:** Simulate a common network issue (e.g., incorrect subnet mask, gateway misconfiguration) and troubleshoot the issue using your understanding of the TCP/IP stack.
20. **Question:**
    - What is the issue you simulated, and how did it affect network communication?
    - How did you diagnose and resolve the issue using packet inspection and knowledge of the TCP/IP layers?
    - What tools and techniques would you recommend for real-world network troubleshooting?

Submission Requirements:

- Submit a report that includes packet captures, command outputs, and explanations for each exercise.
- Include screenshots and highlight key details in the packet captures.

- Provide in-depth analysis of TCP/IP behavior observed during the lab.

Reflection:

Through this lab, students will gain practical experience in dissecting and troubleshooting network traffic using the TCP/IP stack. Understanding how data flows across the network at various layers will help in mastering network fundamentals and preparing for real-world challenges in network administration and cybersecurity.